Rolepilot Hardware Specification Report

Multiple computer and server specifications for running Rolepilot and generating AI assessment tests in under 2 minutes, with practical GPU formats and purchasing guidance.

PROJECT

Rolepilot - FastAPI backend, Next.js frontend, PostgreSQL, and Ollama/Gemma local AI.

TARGET

Generate 3 complete candidate assessment tests from one JD in under 2 minutes.

CURRENT DETECTED MACHINE

NVIDIA GeForce RTX 3050 Laptop GPU, 4GB VRAM, 16 CPU cores, 22GB RAM.

CURRENT OUTCOME

Medium profile: `gemma3:4b`, sequential generation, about 15 minutes.

Main recommendation: choose at least 32GB VRAM for a minimum practical local system, and 48GB+ VRAM for a reliable under-2-minute target.

Prepared for local/server planning. Generated from inspection of the Rolepilot repository and current NVIDIA GPU specifications.

Executive Summary

> The normal backend and frontend app workloads do not need extreme hardware. The expensive part is local AI generation through Ollama. Rolepilot creates 3 tests per job description: Technical, Reasoning, and Problem Solving. Each test generation can ask the model to return a structured JSON test with 5 MCQ questions and 2 open-ended questions.
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> To reach under 2 minutes, the system needs enough GPU memory to keep the model and the parallel request KV cache in VRAM. If the model spills into system RAM, generation speed drops sharply. This is why GPU VRAM is the most important purchasing variable.

Project Findings

Area	What I found	Hardware impact
Backend	Python FastAPI with SQLAlchemy async, PostgreSQL, pytest suite with mocked AI.	Needs solid CPU and RAM, but not a large GPU for normal automated tests.
Frontend	Next.js 15, React 19, Prisma, TypeScript, Tailwind.	Build/type-check speed benefits from modern CPU and NVMe storage.
AI generation	Ollama/Gemma generates 3 tests per JD. The code can dispatch all 3 in parallel when configured.	This is the true bottleneck. Requires strong NVIDIA GPU and enough VRAM.
Current machine	RTX 3050 Laptop GPU, 4GB VRAM, configured as `gemma3:4b` sequential.	Appropriate for development, not for under-2-minute generation.

How To Read The GPU Format

When choosing the GPU, use the exact model name, VRAM amount, memory type, and cooling/form factor. This avoids buying the wrong class of card.

Correct format	Meaning
`1x NVIDIA GeForce RTX 5090 32GB GDDR7`	Consumer/workstation-style GPU with 32GB VRAM. Good minimum practical local choice.
`1x NVIDIA RTX 6000 Ada Generation 48GB GDDR6 ECC`	Professional active-cooled workstation GPU with ECC memory.
`1x NVIDIA L40S 48GB GDDR6 ECC, PCIe, passive cooling`	Data-center/server GPU. Requires proper rack server airflow.
`1x NVIDIA RTX PRO 6000 Blackwell Server Edition 96GB GDDR7 ECC`	High-end Blackwell server GPU with much more VRAM for future models and concurrency.
`1x NVIDIA H100 SXM 80GB HBM3` or `1x NVIDIA H200 141GB`	Enterprise/cloud AI accelerator class. Best performance, highest cost.

Recommended Options

Option 1: Minimum Practical Local Workstation

Budget Serious

CPU	RAM
Modern 16-core desktop CPU, such as Ryzen 9 / Core i9 class.	64GB DDR5 minimum.
STORAGE	**GPU FORMAT**
2TB NVMe Gen4 or Gen5 SSD.	`1x NVIDIA GeForce RTX 5090 32GB GDDR7`
POWER	**OS**
1000W to 1200W quality PSU.	Ubuntu 24.04 LTS preferred.

Why it should be considered: This is the lowest-cost machine I would take seriously for the under-2-minute goal. The 32GB VRAM gives enough headroom for faster local inference than 16GB or 24GB GPUs, and the RTX 5090 has very high compute throughput.

Best use: Local development plus fast generation using `gemma3:12b`, with `OLLAMA_NUM_PARALLEL=3` and `TEST_SEQUENTIAL_GENERATION=false`.

Caution: This is the minimum practical tier, not the safest one. For `gemma3:27b` and three parallel generations, 32GB can be tight depending on context length, quantization, and retries.

Option 2: Recommended Reliable Workstation

Best Balance

CPU	RAM
16 to 24 modern cores.	128GB DDR5, ECC preferred if available.
STORAGE	**GPU FORMAT**
2TB NVMe SSD, plus optional 2TB backup/data SSD.	`1x NVIDIA RTX 6000 Ada Generation 48GB GDDR6 ECC`
POWER	**OS**
1000W+ PSU, workstation-grade cooling.	Ubuntu 24.04 LTS.

Why it should be considered: This is the workstation I would recommend first. The 48GB ECC VRAM gives better reliability and more room for 3 parallel AI generations than 32GB. It is active cooled, so it fits conventional workstations better than passive server GPUs.

Best use: Serious local AI generation, development, and production-style testing on one machine. Suitable for `gemma3:27b` in parallel with more confidence than consumer GPUs.

Caution: It costs more than consumer GeForce cards, but it is easier to trust for long-running professional workloads because of ECC memory and workstation drivers.

Option 3: Recommended Rack Server

Production Server

CPU AMD EPYC or Intel Xeon, 24+ cores.	**RAM** 128GB to 256GB ECC.
STORAGE 2TB enterprise NVMe SSD, with backup/RAID strategy for production.	**GPU FORMAT** `1x NVIDIA L40S 48GB GDDR6 ECC, PCIe, passive cooling`
NETWORK 1GbE minimum, 10GbE preferred for office/server use.	**OS** Ubuntu Server 24.04 LTS.

Why it should be considered: This is the best server-style choice for Rolepilot if the app will run continuously for a team. L40S is made for data-center AI and graphics workloads, has 48GB ECC VRAM, and works well in certified server chassis.

Best use: Deployed Rolepilot server, shared access by multiple admins, background generation, and candidate evaluations.

Caution: L40S is passively cooled. Do not put it in a normal desktop case. It needs a real rack/server chassis with strong front-to-back airflow.

Option 4: Future-Proof AI Server or Workstation

Buy Once

CPU 32+ cores.	**RAM** 256GB ECC.
STORAGE 4TB enterprise NVMe SSD.	**GPU FORMAT** `1x NVIDIA RTX PRO 6000 Blackwell Server Edition 96GB GDDR7 ECC`
ALTERNATIVE `1x NVIDIA RTX PRO 6000 Blackwell Workstation Edition 96GB GDDR7 ECC`	**OS** Ubuntu 24.04 LTS.

Why it should be considered: This is the cleanest single-GPU future-proof option. The 96GB VRAM is far beyond the current app requirement, which means more room for larger models, more concurrent requests, and future changes to prompt size or question count.

Best use: Production AI server, heavier local models, multiple generations at the same time, or future expansion without replacing the GPU quickly.

Caution: This is expensive and may be unnecessary if Rolepilot will only serve one admin generating one JD at a time.

Option 5: Cloud or Enterprise AI Accelerator

Maximum Performance

CPU
32+ vCPU or physical cores.

RAM
256GB+.

STORAGE
2TB+ NVMe, depending on retention and backup policy.

GPU FORMAT
`1x NVIDIA H100 SXM 80GB HBM3`

ALTERNATIVES
`1x NVIDIA H100 NVL 94GB` or `1x NVIDIA H200 141GB`

OS
Provider-supported Linux image with current NVIDIA drivers.

Why it should be considered: Choose this tier if many people will use Rolepilot at once, if candidate submissions arrive concurrently, or if you want the strongest inference performance without worrying about local workstation limits.

Best use: Enterprise deployment, high concurrency, hosted AI service, or future migration from Ollama to a higher-throughput inference server.

Caution: This is usually overkill for a small internal team. Cost can be much higher than the application itself unless the server is shared across multiple AI workloads.

Comparison Matrix

Option	Expected fit for under 2 minutes	Cost level	Reliability	Best buyer
RTX 5090 32GB workstation	Good for `gemma3:12b`; less certain for `gemma3:27b`.	Medium-high	Good, but consumer-class.	Developer or small team.
RTX 6000 Ada 48GB workstation	Strong fit for the target.	High	Very good, ECC workstation GPU.	Professional workstation buyer.
L40S 48GB rack server	Strong fit for the target.	High	Very good in proper server chassis.	Production server buyer.
RTX PRO 6000 Blackwell 96GB	Excellent, with future headroom.	Very high	Excellent.	Long-term AI infrastructure buyer.
H100/H200	Excellent, especially for concurrency.	Enterprise	Excellent.	Cloud/enterprise deployment.

Configuration Notes For Rolepilot

For the under-2-minute goal, the server should run with parallel generation enabled:

- `OLLAMA_NUM_PARALLEL=3`
- `OLLAMA_KEEP_ALIVE=-1`
- `OLLAMA_FLASH_ATTENTION=1`
- `OLLAMA_MAX_LOADED_MODELS=1`
- `TEST_SEQUENTIAL_GENERATION=false`

The current launcher already sets these ideas by hardware profile. The key difference is that the current detected 4GB GPU forces sequential mode. A 32GB to 48GB+ GPU allows the generation path to run the 3 test requests in parallel.

Hardware I Would Avoid For This Goal

- **4GB VRAM GPUs:** fine for development only. Your current RTX 3050 Laptop GPU is in this category.
- **8GB to 16GB VRAM GPUs:** can run smaller models, but not reliable for sub-2-minute full generation.
- **RTX 4090 24GB GDDR6X:** powerful, but still not my first recommendation for the strict target because VRAM is tighter than 32GB/48GB options.
- **CPU-only servers:** acceptable for app hosting, but not for fast local AI generation.

Final Recommendation

If you want the best balance of cost, speed, and reliability, choose **Option 2: NVIDIA RTX 6000 Ada Generation 48GB GDDR6 ECC workstation**. If this will be a real production server in a rack, choose **Option 3: NVIDIA L40S 48GB GDDR6 ECC**.

If budget is tight and this is mostly for one user or a small team, choose **Option 1: RTX 5090 32GB GDDR7**. If you want room for much larger future models and more concurrency, choose **Option 4: RTX PRO 6000 Blackwell 96GB GDDR7 ECC**.

Sources

- Rolepilot repository: `README.md`, `start.sh`, `backend/.env.hardware`, `backend/app/services/test_generation_service.py`.
- NVIDIA GeForce RTX 5090 official page: https://www.nvidia.com/en-us/geforce/graphics-cards/50-series/rtx-5090/
- NVIDIA GeForce RTX 4090 official page: https://www.nvidia.com/en-us/geforce/graphics-cards/40-series/rtx-4090/
- NVIDIA L40S official page: https://www.nvidia.com/en-us/data-center/l40s/
- NVIDIA RTX 6000 Ada official page: https://www.nvidia.com/en-us/products/workstations/rtx-6000/
- NVIDIA RTX PRO 6000 Blackwell official page: https://www.nvidia.com/en-us/products/workstations/professional-desktop-gpus/rtx-pro-6000-family/
- NVIDIA H100 official page: https://www.nvidia.com/en-us/data-center/h100/
- NVIDIA H200 official page: https://www.nvidia.com/en-us/data-center/h200/